Computer Networks

(ENCS352)

LAB FILE SUBMITTED TO

K. R. Mangalam University

FOR

Bachelor of Technology

in

Computer Science and Engineering

SUBMITTED BY

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Assignment 2: Packet Flow Visualization Using Simulation Mode

Experiment Objectives:-

Students will use Packet Tracer Simulation Mode to observe how packets move between devices,how switches learn MAC addresses, and how ICMP echo request/reply travels through a small network.

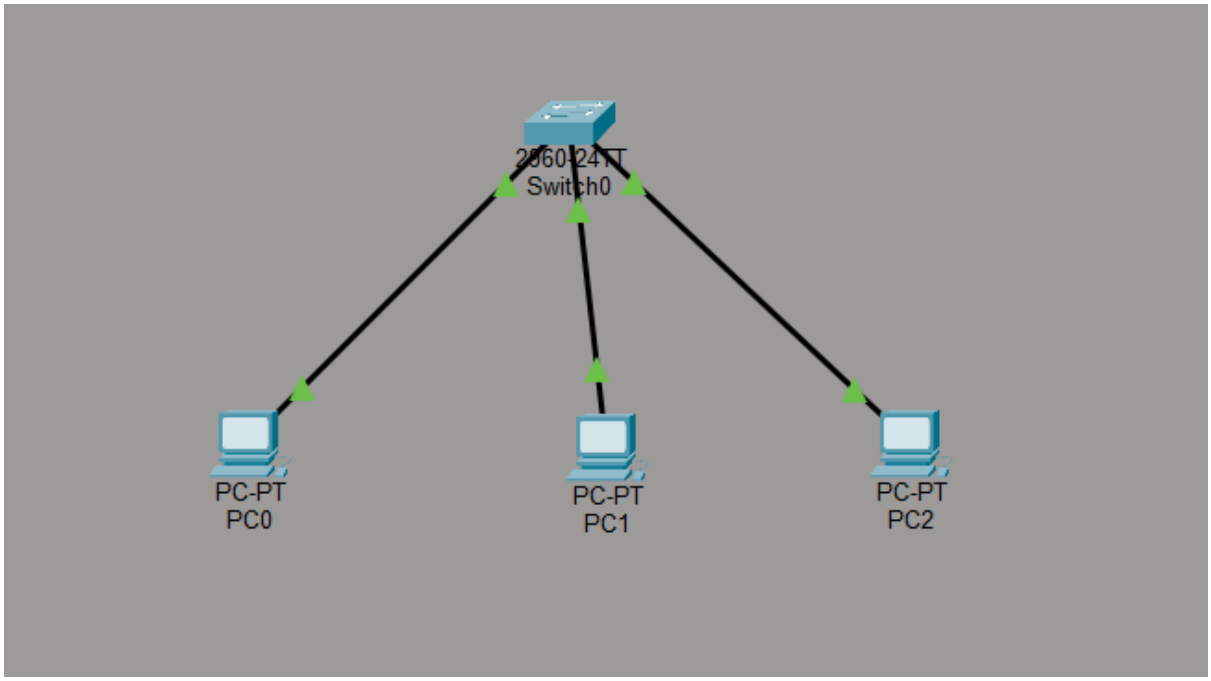
Learning Outcomes:-

- Use Simulation Mode effectively
- Observe ARP + ICMP behavior
- Understand switch MAC learning behavior
- Interpret packet event list timeline

Concepts Used:-

- Simulation Mode
- ICMP ping
- ARP resolution
- Switch MAC address table learning

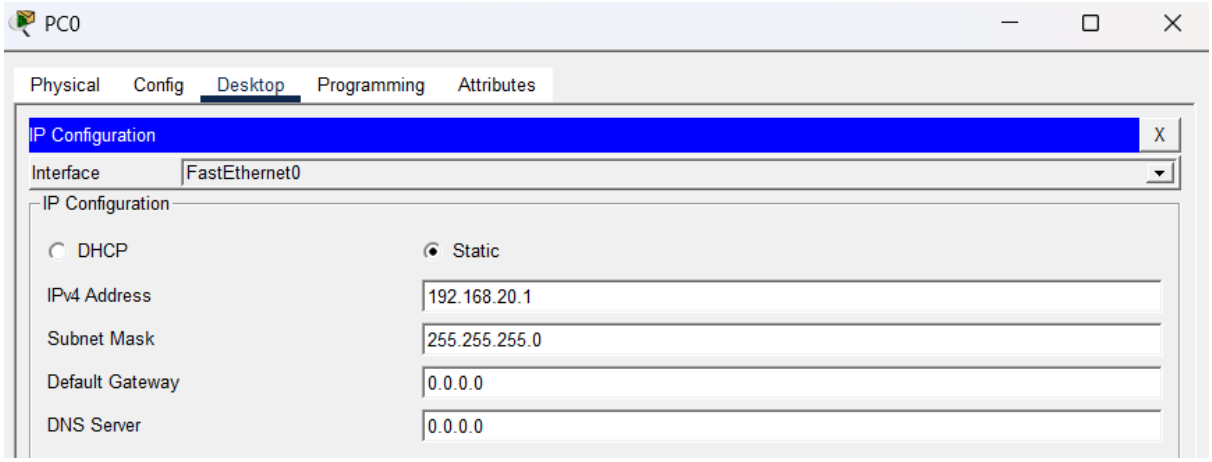
Task 1: Build a Simple LAN

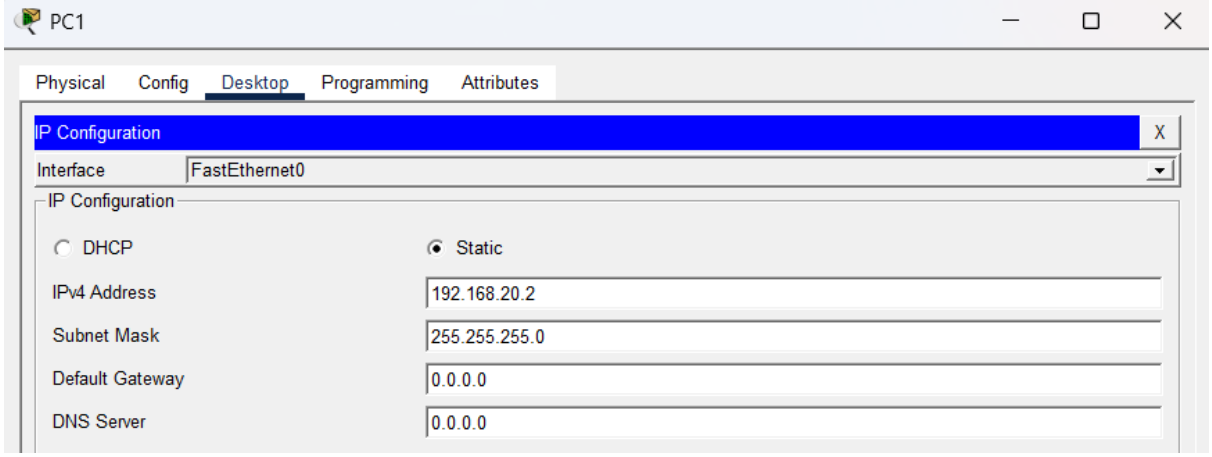


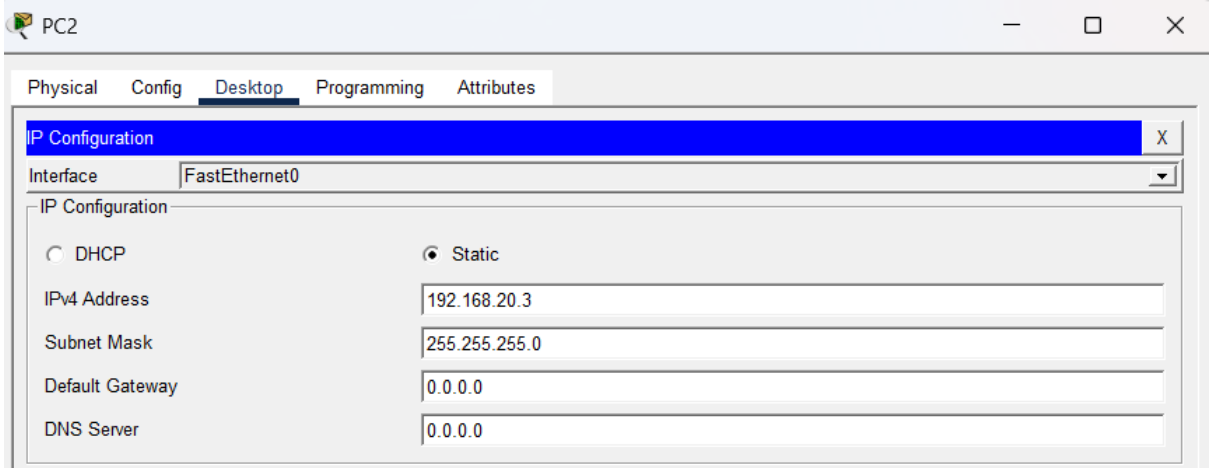
Configuration:-

- 1 Switch (2960)
- 3 PCs connected to switch
- IP addresses assigned:

PC	IP Address	Subnet Mask
PC0	192.168.20.1	255.255.255.0
PC1	192.168.20.2	255.255.255.0
PC2	192.168.20.3	255.255.255.0

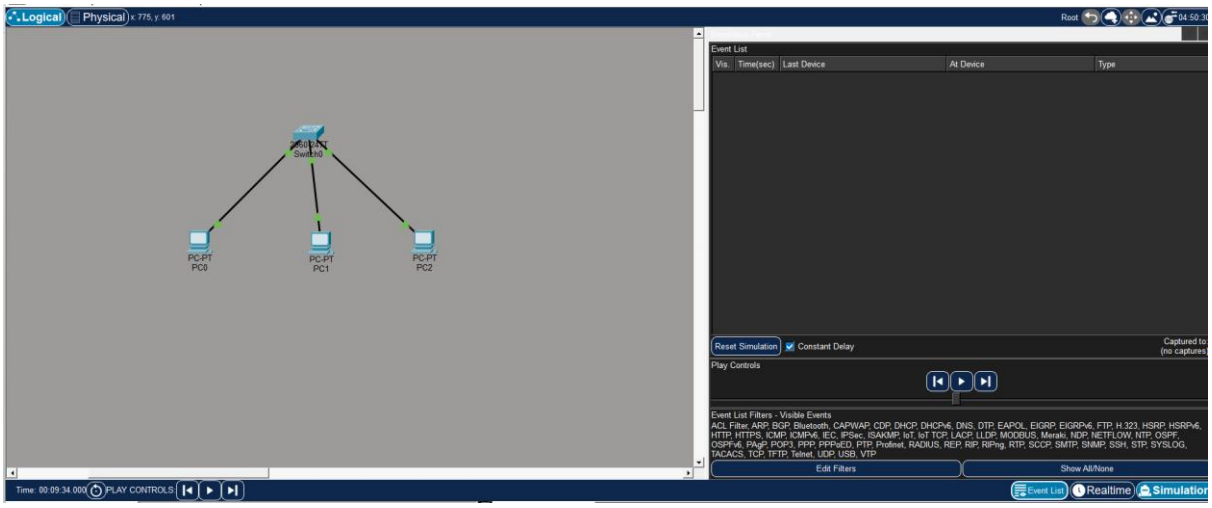






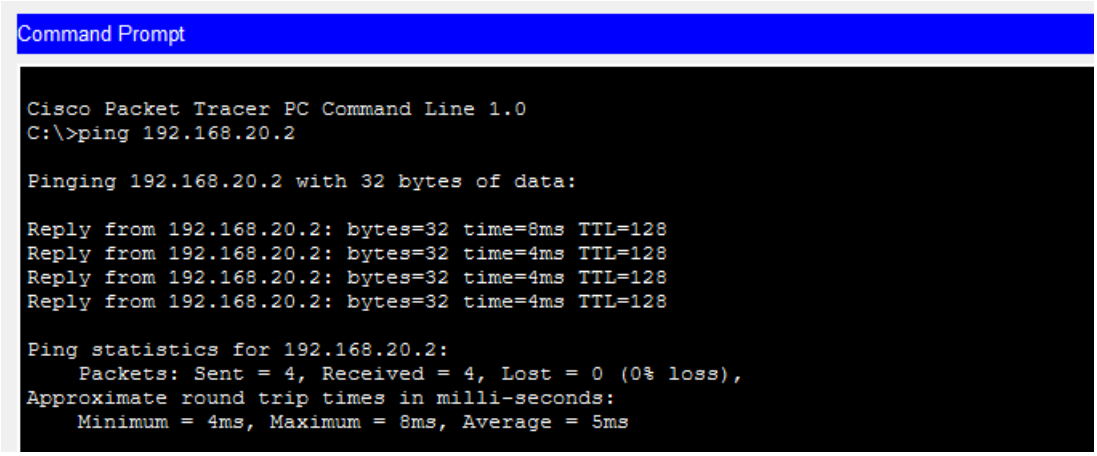
Task 2: First Ping Observation

- Simulation Mode was enabled..



- The Event List filter was set to show only ARP and ICMP.

- A ping was issued from PC1 to PC2 via the Command Prompt: ping 192.168.20.2

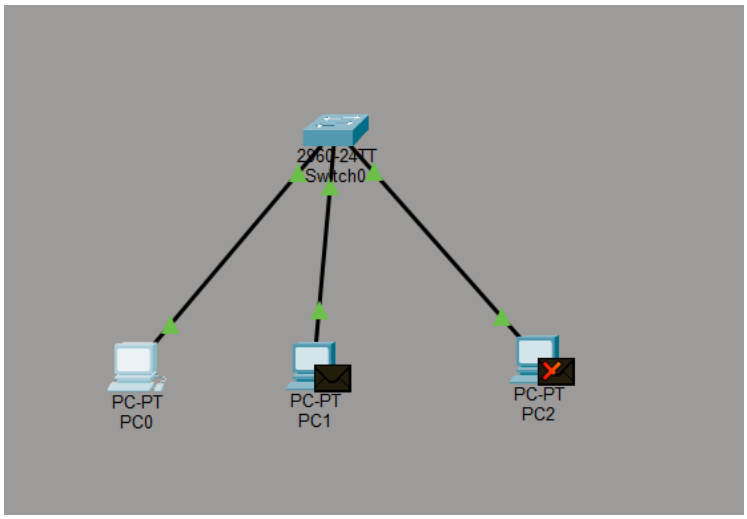


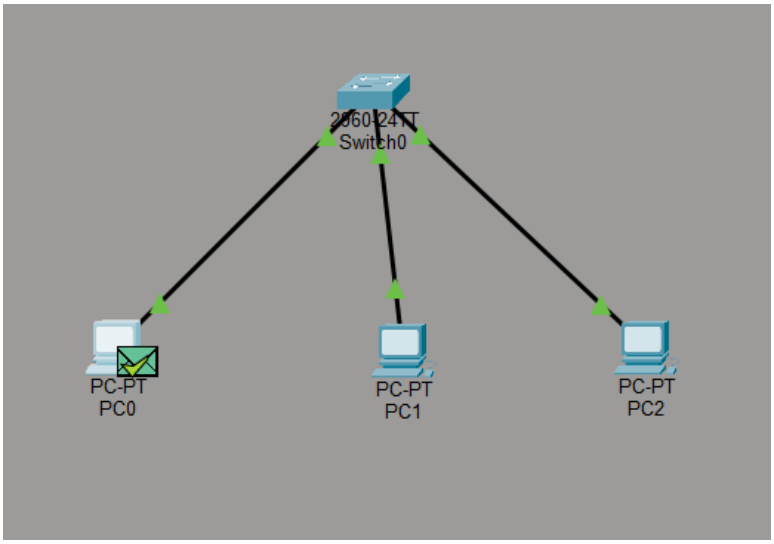
Observed Event Sequence:-

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.000	--	PC0	ARP
	0.001	PC0	Switch0	ARP
	0.002	Switch0	PC1	ARP
	0.002	Switch0	PC2	ARP
	0.003	PC1	Switch0	ARP
	0.004	Switch0	PC0	ARP
	0.004	--	PC0	ICMP
	0.005	PC0	Switch0	ICMP
	0.006	Switch0	PC1	ICMP
	0.007	PC1	Switch0	ICMP
	0.008	Switch0	PC0	ICMP
	1.010	--	PC0	ICMP
	1.011	PC0	Switch0	ICMP
	1.012	Switch0	PC1	ICMP
	1.013	PC1	Switch0	ICMP
	1.014	Switch0	PC0	ICMP
	2.018	--	PC0	ICMP
	2.019	PC0	Switch0	ICMP
	2.020	Switch0	PC1	ICMP
	2.021	PC1	Switch0	ICMP





Task 3: Second Ping Observation

Without clearing the ARP cache, a second ping was issued immediately: ping 192.168.20.2. The simulation was re-run from the beginning of the event list for the new set of packets.

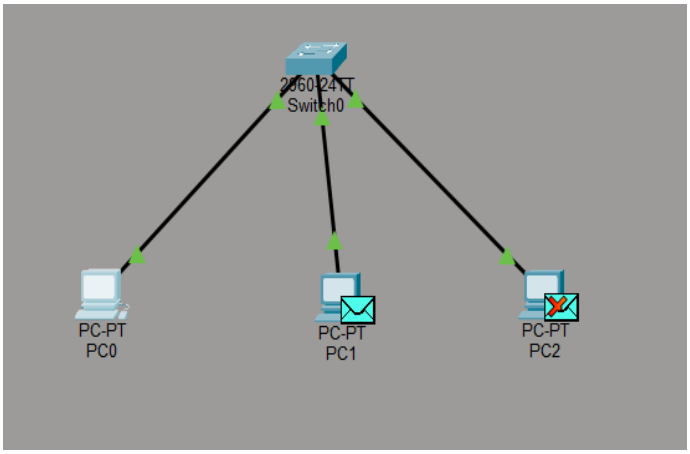
```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms
```

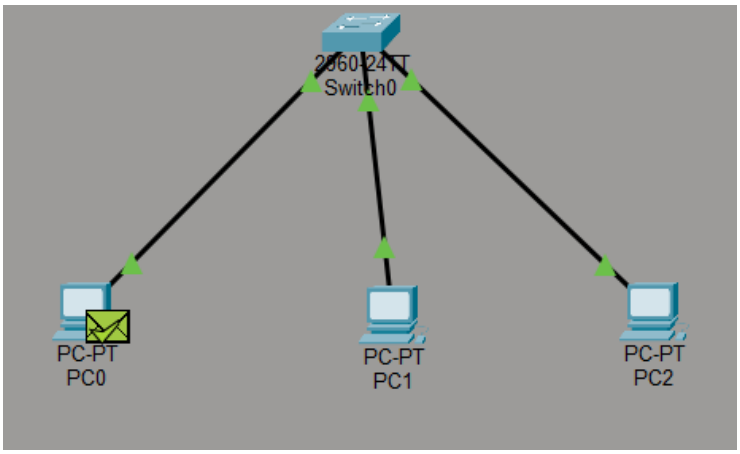
Observed Event Sequence:-

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	PC1	ICMP
	0.002	Switch0	PC2	ICMP
	0.003	PC1	Switch0	ICMP
	0.004	Switch0	PC0	ICMP
	1.006	--	PC0	ICMP
	1.007	PC0	Switch0	ICMP
	1.008	Switch0	PC1	ICMP
	1.009	PC1	Switch0	ICMP
	1.010	Switch0	PC0	ICMP
	2.013	--	PC0	ICMP
	2.014	PC0	Switch0	ICMP
	2.015	Switch0	PC1	ICMP
	2.016	PC1	Switch0	ICMP
	2.017	Switch0	PC0	ICMP
	3.021	--	PC0	ICMP
	3.022	PC0	Switch0	ICMP





Task 4: MAC Table Check

```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    00e0.8fe5.0261    DYNAMIC     Fa0/1
   1    00e0.b0a8.a266    DYNAMIC     Fa0/2
Switch#
```

- **First Ping vs. Second Ping – Comparison**

Feature	First Ping	Second Ping
ARP Used	Yes	No
Speed	Slow	Fast
Packets Count	More	Less
Reason	MAC unknown	MAC cached